Assignment 6

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JOIN exercises

1. List employees and their managers.

   SELECT e.name AS “Employee”, m.name AS “Manager”
   FROM employees e
   INNER JOIN employees m ON e.manager_id=m.id;
   (use LEFT JOIN to include employees with NULL depts)

“Employee”	“Manager”
Charlotte	Emma
Ella	William
Scarlett	Olivia
Ava	Ethan
Lucas	Ava
Grace	Ethan
Charlotte	Ava
David	Olivia

174 rows in set (0.121 sec)

2. List departments and their managers.

   SELECT d.department_name AS Department, e.name AS Manager
   FROM departments d
   INNER JOIN employees e ON d.manager_id=e.id;
   (use LEFT JOIN to include employees with NULL managers)

Department	Manager
Assembly	James
Quality Control	Maria
Machining	Robert
Logistics	Emma
Maintenance	Liam
R&D	Olivia
Procurement	Noah
Health & Safety	Ava
Packaging	William
Inventory	Sophia
Welding	Ethan
Plant Management	Isabella

12 rows in set (0.001 sec)

3. List department that each employee works in.
   SELECT e.name AS Employee, d.department_name AS Dept
   FROM employees e
   INNER JOIN departments d ON e.department_id=d.id;

Employee	Dept
James	Assembly
Scarlett	Assembly
Olivia	Assembly
Charlotte	Plant Management
Charlotte	Plant Management
Grace	Plant Management

176 rows in set (0.032 sec)

4. List employees with no manager.
   SELECT name
   FROM employees
   WHERE manager_id IS NULL;

```
+----------+
| name     |
+----------+
| James    |
| Maria    |
| Robert   |
| Emma     |
| Liam     |
| Olivia   |
| Noah     |
| Ava      |
| William  |
| Sophia   |
| Ethan    |
| Isabella |
| Omar     |
| Sherif   |
+----------+
14 rows in set (0.006 sec)
```

5. List employees with no department.

```
SELECT name
FROM employees
WHERE department_id IS NULL;
```

```
+-----------+
| name      |
+-----------+
| Mia       |
| Ava       |
| Oliver    |
| Omar      |
| Aiden     |
| Olivia    |
| Logan     |
| Mateo     |
| Sherif    |
| Sebastian |
| Ava       |
| Charlotte |
+-----------+
12 rows in set (0.001 sec)
```

6. Count employees in each department, including those without departments.
   SELECT d.department_name, COUNT(e.id) AS Employee_Count
   FROM departments d
   LEFT JOIN employees e ON d.id=e.department_id
   GROUP BY d.department_name;

department_name	Employee_Count
Assembly	16
Health & Safety	24
Inventory	16
Logistics	13
Machining	13
Maintenance	10
Packaging	11
Plant Management	16
Procurement	13
Quality Control	12
R&D	15
Welding	17

12 rows in set (0.001 sec)

7. List name, salary, and department of all managers.
   SELECT e.name, e.salary, d.department_name
   FROM employees e
   INNER JOIN departments d ON e.id=d.manager_id;

name	salary	department_name
James	95000	Assembly
Maria	92000	Quality Control
Robert	88000	Machining
Emma	91000	Logistics
Liam	85000	Maintenance
Olivia	110000	R&D
Noah	78000	Procurement
Ava	82000	Health & Safety
William	75000	Packaging
Sophia	84000	Inventory
Ethan	89000	Welding
Isabella	125000	Plant Management

12 rows in set (0.001 sec)

8. List employees with salaries higher than their managers.

```
SELECT e.name AS Employee, e.salary, m.name AS Manager, m.salary AS Manager_Salary
FROM employees e
INNER JOIN employees m ON e.manager_id=m.id
WHERE e.salary>m.salary;
```

Employee	salary	Manager	Manager_Salary
Ella	97000	James	95000
Scarlett	77000	William	75000

2 rows in set (0.032 sec)